

XR-OOM: MiXed Reality driving simulation with real cars for research and design

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Figure 1: XR-OOM enables experimental studies where participants encounter virtual objects (cones) while driving a real car.

ABSTRACT

High-fidelity driving simulators can act as testbeds for designing in-vehicle interfaces or validating the safety of novel driver assistance features. In this system paper, we develop and validate the safety of a mixed reality driving simulator system that enables us to superimpose virtual objects and events into the view of participants engaging in real-world driving in unmodified vehicles. To this end, we have validated the mixed reality system for basic driver cockpit and low-speed driving tasks, comparing the use of the system with non-headset and with the headset driving conditions, to ensure that participants behave and perform similarly using this system as they would otherwise. This paper outlines the operational procedures and protocols for using such systems for cockpit tasks (like using the parking brake, reading the instrument panel, and turn signaling) as well as basic low-speed driving exercises (such as steering around corners, weaving around obstacles, and stopping at a fixed line) in ways that are safe, effective, and lead to accurate, repeatable

data collection about behavioral responses in real-world driving tasks.

CCS CONCEPTS

• **Human-centered computing** → **User studies; Mixed / augmented reality.**

KEYWORDS

XR, mixed reality, driving simulation, automotive, user studies, design, safety

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1 INTRODUCTION

Automotive designers and human factors engineers use driving simulation environments to test the usability, safety, and driver performance impacts of interfaces and interactions. Driving simulators can help recreate scenarios that may rarely occur in the world or would pose an unreasonable risk to study participants to stage. By controlling experimental procedures and scenarios, driving simulators can help recreate the same factors and events within and between experimental conditions. In addition, driving simulators

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can act as a testbed for designers, allowing them to rapidly develop and prototype for specific situations and environments and see how people might behave in those contexts.

It is common to use a mix of high and low-fidelity test environments throughout the development of vehicular systems [15]. Current-day fixed-base simulators in laboratories usually feature screen-based instrument panels and center consoles to enable the rapid development of user interfaces. They are authored in graphical simulation environments (such as Unity or Unreal) to make it easy to create contexts and scenarios. The level of fidelity of driving simulators has been found to have effects on drivers, such as the presence of a realistic cabin leading to conservative driving styles as well as reduced motion sickness [8]. Test-track environments are more challenging to develop interfaces and scenarios for than simulators but feature more realistic driving experiences such as vibration and physical motion. More naturalistic track or real road studies also help clarify the impact of external factors that designers might otherwise be blindsided by.

We seek to combine the strength of immersive simulation environments and real-world testing environments through the use of mixed reality (XR) driving simulation. In this paper, we validate the safety and viability of this combination towards the goal of using mixed reality to simulate driving scenarios with advanced driver assistant systems. In Section 3, we discuss the design requirements to make in-vehicle XR systems useful for driving simulation experiments in research and design. Next, in Section 3, we describe the mixed reality system that we built to operate driving simulation experiments in vehicles. In Section 4, we outline a safe and effective operational procedures and protocol we used to validate the systems. We validated the mixed reality system for basic driver cockpit and low-speed driving tasks, comparing the use of the system with non-headset and headset-only driving conditions, to ensure that participants behave and perform similarly using this system as they would otherwise.

This work contributes a critical step towards safer, less expensive development environments for designing and testing future vehicle interfaces. Using a mixed reality headset in a real-world test vehicle, simulated events and interfaces can be tested while drivers operate and experience being in an actual car. Such an approach would allow for higher-fidelity experiments where simulation efforts can be focused on interfaces, situations, and designs at hand - reducing efforts spent on replication of a realistic driving experience. We are sharing the specification and validation of our system to make the approach accessible to the broader automotive research community.

2 RELATED WORK

Recent research, as well as promotional materials at industrial expos, have shown the viability of using mixed reality for driving simulation [9, 18, 54, 58, 64]. Such systems use “XR,” (which variably stands for miXed Reality or eXtended Reality [22]) to “pass-through” video of the surrounding environment while inserting virtual objects and virtual interface overlays on the environment so that they appear to be in the physical world. This is a promising demonstration of a technology long-awaited by human-factors researchers [26]. However, to make such systems usable for experimental research

purposes, the methodology and protocol for experiments using this equipment need to be validated and demonstrated to be safe to run.

To better understand how this mixed-reality in-vehicle driving simulation can support automotive research, we review the advances in driving simulation technologies and discuss research that points to key requirements needed to make these technologies safe, valid, and repeatable for automotive studies.

2.1 High-immersion Driving Simulation

Driving simulation takes many different forms, from rudimentary PC-based software simulators (such as City Car Sim [13] where people engage in driving-like tasks using gaming wheels and pedals in front of a computer screen) to highly immersive systems that use tremendous mechanical systems to replicate the inertial movement and forces felt in actual driving (like with Toyota’s high-performance simulator [47] or Ford’s VIRTTEX Simulator [23]). For research purposes, the level of fidelity in visual scene construction or physical motion does not always need to be higher to be better; what is desirable may vary depending on the experiments the simulator is supporting. For example, a simulator used for human-factors experiments with people that have medical impairments (such as [27, 51]) might have relatively low physical motion fidelity but need to very accurately capture the reaction time of the participant to a visual stimulus. In contrast, a system to test motor responses, say, of older drivers, might need far better control and dynamics models to simulate realistic responses properly. For human-machine interaction studies, the point of driving simulators is to help designers understand users’ behavioral responses to displays or notification systems in the car, as well as to events in or outside the car [2].

2.2 Driving Simulation for Autonomy and Driver Assistance

Autonomous vehicles bring the need for new models for visual display systems, control interfaces, audio alerts, and interaction, and the need to test these novel systems [1, 16]. Much of current day driving simulation studies around automation have focused on “transitions of control” [21, 43], but the need to reinvent the interfaces and interactions for autonomous vehicles also makes the use of simulators as design test beds more important. Some researchers have developed a more design-like and improvisational approach to developing interactions with automation [46, 59]. Others have contemplated how, for example, to prototype interactions between the vehicle and brought-in devices such as phones and tablets [3], which people are likely to be focused on if they do not have to drive.

2.3 Mixed Reality Simulation Systems

The VR-OOM system developed by Goedicke et al. employed a virtual reality headset in a wizard-driven vehicle with a false haptic steering wheel to create simulated autonomous driving experiences that combined virtual environments and events with the physical sensations of being in a real car [19, 65]. This work (and the earlier RRADS protocol [5]) argued that non-motion-based driving simulations lose immersivity when used for autonomous driving scenarios because participants in autonomous car studies often do not need to attend to the simulated environment outside the vehicle.

Therefore, they argue, face validity is lowered. These researchers felt they could better produce ecologically valid responses in participant behavior by simulating scenarios and interfaces in an actual vehicle and devising ways to register commonalities between the physical and virtual space.

Yeo et al. prospectively explored the visual and motion fidelity of in-vehicle mixed reality systems using their MAXIM driving simulator [67]. Comparing six different combinations of visual and motion fidelity for user testing of self-driving cars, they found that this combination of real-world mixed reality and real-vehicle produced the highest degree of fidelity and presence of the setups they tested. This early work suggests that on-road mixed reality driving simulators like XR-OOM would better enable vehicle manufacturers to test real vehicle systems with people than a virtual reality-based on-road driving simulator.

2.4 Experimental Range

Another critical factor in developing driving simulations for human-machine interaction design is the trade-off between design range and fidelity. Virtual driving simulation environments can simulate a wide range of driving interfaces, control scenarios, and driving contexts; driving simulation environments that take advantage of current-day gaming and 3D authoring tools have very few limits in this regard.

Mixed reality driving simulations are more constrained by the vehicle and driving environment features but have far more flexibility and range than more traditional methods. Traditional in-lab driving simulators rely on the familiarity of the conventional car environment to produce realistic responses from participants [24]. However, simulators like RRADS [5], VR-OOM [19], or MAXIM [67] may be better suited to test and monitor participant responses to completely novel vehicle interfaces and experiences because they happen on-road providing additional familiarity of the actual driving environment to support the face validity of the driving task. While the developers of these simulators were interested in the physical, bodily, environmental, and social reality that is the basis of a realistic experience [32], one of the key benefits of such systems to automotive designers is the ability to perform controlled human participant studies with simulated events in actual cars. The addition of mixed reality makes it possible to test novel prototypes of existing systems, such as “Guardian” [55] Advanced Driver Assistance Systems, which might break with different amounts of force, closer or further from potential obstacles.

2.5 The Need for Hybrid Virtual/Physical Simulation

From prior research, it is clear that mixed reality on-road driving simulators offer benefits, such as realistic driving experiences and a wide range of flexibility for prototyping virtual interfaces and situations. Prototyping interfaces in XR could reduce costs and allow for rapid iterations. In contrast, simulated environmental conditions (e.g., a goose stopping in front of a car) could allow for realistic testing of complex scenarios on the road in their most realistic manner (without using a real bird).

New XR on-road simulation systems, such as VR-OOM [20], MAXIM [67] and ARCAR [18] enable a wider variety of driving

scenarios to be tested in a safe and controlled manner. Offering XR in-car simulation as an additional research platform can broaden the range of scenarios that can be tested, ultimately supporting efforts to increase the safety of cars.

3 SYSTEM REQUIREMENTS

Our system, XR-OOM, establishes the possibility of using mixed reality for experimental driving simulation research. (See Figure 2.) It is built using the Varjo XR-1 Mixed Reality headset, and Unity simulation environment, which previous researchers demonstrated could be usable for driving simulation in a moving vehicle [18]; XR-OOM integrates and validates these into a usable driving simulation system. This combination of software and hardware is the only practicable implementation of XR-OOM at submission time.

This section motivates the system’s setup based on prior immersive XR and driving simulation systems. These requirements address the usability of such a system and the participants’ safety, well being, immersion, and comfort.

This summary of requirements should help replication attempts to evaluate other potential software and hardware to be used in such systems, as technology progresses and other solutions become viable.

3.1 Headset - Display

XR-OOM participants see the real world and virtual objects through a video pass-through XR headset, which superimposes generated visuals on top of video capture of the real world. Since this is the primary source of information for the participant to drive the vehicle, the headset display is also the technological bottleneck. It needs to be able to faithfully project the real world and add virtual objects accurately. The requirements for the display and cameras are as follows:

3.1.1 Visual Resolution. The visual resolution influences the level of immersion of the participant and their ability to acquire situational awareness. Technically, this is bound by both the screen’s resolution and the cameras used to capture the environment.

One crucial functional requirement is the participant’s ability to read text, for example, on navigation displays or the instrument panel. Text readability is challenging in the virtual reality context because it requires very high display resolution [25, 38]. Prior research by Dingler et al. indicates that the resolution needed for text readability in VR is a function of the text size and render distance [14]. Curevo et al. found that “life-like” virtual reality requires a resolution of 60 pixels per degree for 20/20 vision [12].

3.1.2 Field-of-view. Whereas visual resolution concerns the number of pixels per angular degree-of-view, the field-of-view (FoV) requirement is a function of one’s peripheral perceptions. Van Erp and Padmos argue that FoV is critical to lateral control tasks, leading to a stronger impact on driving performance than visual resolution or latency [63]. Driving performance was better in drivers with a 100° FoV compared to those provided with a 50° FoV. Hu et al. note that the human eye has a 210°×150° (i.e., diagonal 200°) [12] FoV. Most commercial VR headsets only achieve a maximum 150° of

FoV¹. The most critical zone, however, is the 60° around the center of the view [29]. Similar to the visual resolution to resolve details, the FoV requirement depends to a certain extent on the experimental task the simulator is supporting. Any tests which draw on the participant's situational awareness or response to out-of-vehicle cues or events would need a broad FoV. Tasks that take place in a slow traffic environment or center on in-vehicle interaction may use just a subset of a typical human FoV.

3.1.3 Pass-through latency. Pass-through latency describes the combined latency of capturing the real-world with a camera and projecting it onto the screen with all processing steps in the middle. Research on the combined latency effects on driving performance in simulated teleoperation of vehicles indicates a delay of 300 ms degrades driving performance [50]. Related research looking at network-induced latency jitter indicates that jitter is a stronger predictor of negative remote driving performance, but also found a performance threshold at the 350 ms range [40]. Studies on networked multiplayer video games, in particular racing games, set the acceptable latency range at 50 ms [10, 52]. With longer delays between the participant control and visual response, the perception and performance of game players start to degrade noticeably.

Mixed and Virtual Reality applications, however, generally must meet stricter latency requirements than those we found in remote operation and multi-player gaming [12]. Mania et al. identified the Just Noticeable Difference point to be at about 15 ms with up to 20 ms being referenced as the maximum amount of delay before performance and immersions start to be impacted [41]. Hu et al. indicate that the VR interaction latency should be less than 20 ms to avoid motion sickness and discomfort [29]. Hence we believe that systems that participants can operate in without feeling motion sickness or discomfort should be reasonably good at supporting driving operations and on-road coordination with other drivers.

3.2 Tracking and Localization

The system needs to solve a nested tracking problem for the simulator to render objects relative to the car and relative to the outside world.² We use the SmartTrack3 from ART [36] to track the participant's head position, and a ZED2 Stereo Depth Camera [61] mounted on the front bumper of the research vehicle for in-world location (see Figure 2).

3.2.1 In-car localization. The head pose of the participant needs to be tracked relative to the vehicle cabin to accurately render the in- and out-of vehicle scenes from the driver's perspective. This makes it so that virtual objects within the car—for example, novel instrument panel displays or in-world study questionnaires or instructions—can appear to be fixed even if the participant moves their head.

3.2.2 In-world localization. Localizing the car's position and movement is key to properly anchoring virtual objects into the real-world

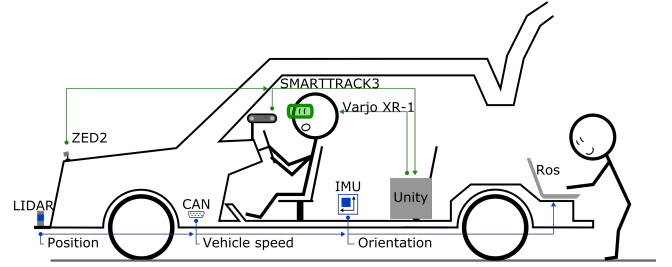


Figure 2: Layout of the XR-OOM system components and validation equipment in research vehicle.

scene outside the vehicle. Combining in-car localization with vehicle localization gives us a continuous estimate of the headset's location and orientation within the vehicle and the world.

3.3 Additional Hardware, Software and Instrumentation

3.3.1 Hardware Configuration. Details of our custom hardware implementation are detailed in the Appendix: A.1. System diagrams, code, and simulator data for XR-OOM are on Github (see Acknowledgements). The details of the required system configuration have not been previously documented by earlier publications about proof-of-concept systems. They are necessary to any researchers looking to use such a system to perform research experiments with [18].

3.3.2 Custom Software. We developed software to render virtual objects in the real world outside the car and to obscure those objects when the physical car interior would have occluded the physical counterparts of these virtual objects. This required synchronization between the location of the participant's head in the car, the location of the car in the space, and the location of the virtual objects in the space. This code is available on GitHub (see Acknowledgements).

3.3.3 Research Instrumentation. For experimental research purposes, the driving simulation system needs to be able to capture the environment, the participant's perspective, the behavioral response of the participant, the vehicular response from the car—any aspects of the uncontrolled field environment which might influence the study outcomes. We use the eye tracker of the XR-1, a 360° activity camera mounted under the rear-view mirror, and an external activity camera stationed outside the vehicle to capture the surroundings.

4 SYSTEM VALIDATION

We perform a system validation test to evaluate our XR-OOM system to benchmark the external validity of simulated activity; this validation measures how similarly participants respond to real-world vs. virtual objects while using XR-OOM.

The following system validation was focused on operational issues involved in running mixed reality in-vehicle driving simulations. In addition, we evaluate the safety and comfort when using the system, effects on driving performance, and usefulness in understanding and recording driver interactions.

¹The Headset with the highest FoV that we could find is the Vision-8k-x with an FoV of 200° <https://pimax.com/product/vision-8k-x/> Retrieved January 8th, 2022

²For example, in [45] blog post, the Hololens team notes recently being able to perform similar things for boat motion, and specifically mentions that such performance is not yet possible for rendering virtual objects in moving systems like cars.

4.1 Pilot Test

In pre-validation pilot testing, we were able to demonstrate that the systems graphic rendering works on an experiential level. The registration between virtual objects and the physically projected world was convincing, such that it matched the physical sensations.

The pilot test involved four different runs with increasing technical complexity. The first pilot had the participant in the passenger seat and reported their experience. The following three pilots then implemented more and more of the final study design, gradually increasing complexity—the pilot participants were members of the same University lab that were not actively working on this project.

The main takeaways from the pilot tests were that 1) it was possible to acquire situational awareness while using the XR-headset, 2) it is possible to operate the vehicle in a low-speed environment safely while using the XR-headset, 3) backward driving required more work, as navigating backward without properly working mirrors that include virtual objects was not realistic. We also learned more minor details around the study execution and how to instruct participants best using the setup.

Unlike this pilot test, the subsequent validation test aimed to ensure that the system works for a broader audience of participants and verify that the system is practical for use for driving simulation studies.

4.2 Sensor and Localization Validation

To validate the XR-OOM system's ability to track and localize the research vehicle as it moves through the real-world study environment, we set up a secondary trajectory tracking system to benchmark XR-OOM's tracking and localization capabilities. This system also enabled us to track vehicle motion through all study conditions, including the control condition where the XR-OOM system was not used.

The trajectory was computed using the Simultaneous Localization and Mapping (SLAM) solution from Koide et al. [39] running in Robot Operating System (ROS) [56] with data from a 360°1024 beams with 20 HZ (Ouster OS1-64), an IMU sampling at about 1KHz (Xsense MTi-300), and odometry data (vehicle speed) from the vehicle's CAN bus.

4.3 Validation of Simulator as Testing Platform

To ensure the system's capability to capture driving performance and participant response, and to understand the impact of the system on both these factors, we conducted a validation test to ensure that the system meets the criteria for being used to conduct user studies typical of those that human-machine interaction designers and researchers run. The criteria that we investigate are the following:

- (1) Does using the video pass-through headset allow drivers to conduct cockpit tasks?
- (2) Does the video pass-through headset allow participants to maintain a sufficient driving performance and experience?
- (3) Are virtual objects simulated believably enough to provide valid predictions of how drivers will respond to physical objects in driving?

- (4) Does the system support data recording to the extent that is required to run interaction user studies?

Moreover, all of these functionality checks needed to occur while the vehicle was being safely operated, so this was also a validation of the safety of the system for this use case.

For this validation, we had participants perform cockpit tasks and driving tasks with and without the XR headset. Our test version of the XR-OOM platform was implemented in a Toyota Prius V, used for the validation tests in a restricted testing area in a parking lot, with permission from the municipal lot owners.

4.4 Participants

As our test was focused on system validation rather than experimental inquiry, the validation sample of participants was drawn from our university through convenience sampling. One of the participants' recording was incomplete and hence dropped from the analysis. The recording for one of the participants is incomplete, hence we removed the participant from subsequent analysis. The resulting sample of 10 participants covered the following estimated demographic characteristics: age (20-30), gender (5 male, 5 female, 0 non-binary), nationality (India, United States). Two participants reported only having driven right-hand drive vehicles before this study.

4.5 Method

4.5.1 Experimental Design. Our experiment had three conditions: *Condition A. No headset* Participants would not wear a headset when conducting tasks and act as a type of control against which the other conditions can be compared against.

Condition B. Headset with video pass-through only Participants would wear a headset that would show a pass-through video of the environment but which included no virtual objects. This condition isolates the impact of wearing and using the XR-1.

Condition C. Headset with video pass-through and virtual objects Participants would wear a headset that would show the same view as condition B but with virtual objects overlaid. This condition evaluates the participant's ability to see and react to virtual objects.

This was a within-subject study design; all the participants experienced all conditions. We used a Latin Square ordering method to counterbalance the order of the headset conditions for each participant.

4.5.2 Cockpit Tasks. Part of the experiment was to perform several non-driving tasks in the cockpit. These cockpit tasks were designed to test the participant's ability to operate the cockpit interfaces of the car. The cockpit tasks were done two times, once without and once with the headset. They were added to the beginning of the no-headset condition(A) and the first condition with a headset (either condition B or C).

We focus our cockpit task validation on stationary tasks that fit into the category of what Bubb et al. call "secondary tasks," [7] which involve communicating with other traffic participants or reacting to the environment changes. Based on the literature [17, 34, 42, 62], we developed the following list of stationary validation tasks: **Turn on vehicle, Adjust seat, Fasten seat belt, Adjust mirrors, Use turn signals, Control headlights, Control**

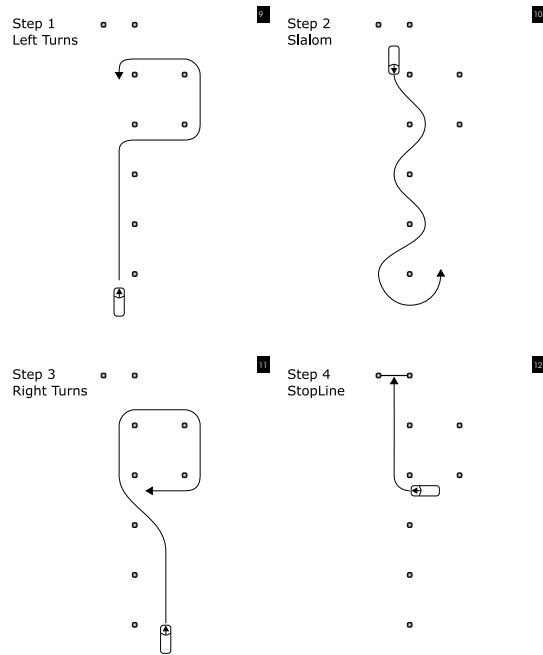


Figure 3: The cockpit tasks the participants drove.

wipers, Control hazard lights, Using the parking brake, Verbally explain which dashboard lights are visible. The version presented to the participant can be found in Appendix: A.2.

This is an important validation step for a simulator as there are many cases in which information from the dashboard or road needs to be read and reacted to by the driver. A simulator that severely limits the participant's ability to resolve the world around them would be significantly limited.

4.5.3 Driving Tasks. Following the stationary cockpit tasks, we evaluate the mixed-reality in-vehicle driving tasks around the category of Bubb et al.'s "primary tasks" [7] which are to keep the vehicle in a planned path and include navigation, handling the road situation in front, and low level lateral and longitudinal control.

The use of a video pass-through headset can introduce driving impairments similar to those of people of advanced age who experience degradation of driving abilities [4]. Their driving impairments could potentially affect the participant in a multitude of ways. These could be cognitive impairments impacting, e.g., their visuospatial ability, speed, and reaction time; vision impairments such as visual acuity, visual fields, color vision, depth perception, contrast sensitivity, and glare; and physical impairments such as limited neck rotation and trunk rotation. Our validation activities aim to ensure that the impairment due to the video headset does not unduly affect primary driving capability.

Drawing upon driving task literature [35, 44, 57], as well as resources for people learning to drive in the US[11], we defined four driving tasks: **Left Turns, Slalom, Right Turns, Stop Line** (see Figure: 3). The tasks took place in a parcourse marked using

traffic cones. In the no-headset (A) and a headset with video pass-through condition (B), the cones were physical cones placed on the road at 8-meter intervals according to a pre-defined pattern. In the headset with virtual objects condition (C), the physical cones were removed from the lot, and replaced with virtual cones displayed within the headset, set out in the same pattern as the physical cones.

These first evaluation steps evaluate the participant's capability to safely operate the vehicle and perform basic driving and cockpit tasks, forming a driving simulator's baseline requirements.

4.5.4 Procedure. Each participant's engagement took about an hour in total. The participant was first asked to give informed consent for the study and then was transported from campus to the testing site. The participant was informed that they could stop the experiment at any time for any reason. They were also informed that they should let us know if they feel motion sick or unwell in any other way. We asked them to drive slowly and that, in the unlikely event of a failure of the mixed reality system, they should stop the vehicle by stepping on the brake. The participants performed all the study tasks as ordered by the study's Latin Square design. After the driving tasks, the participants answered simulator sickness and subjective experience questions. After the experiment, the experimenters would transport the participant back to campus.

4.5.5 Measures. We used a variety of observed, objective, and self-assessment measures to understand the XR-OOM system's impact on people's driving.

Observed measures. Video of participant performance on the cockpit and driving tasks was observed and rated by researchers post-facto.

Objective measures. The driving performance was measured using post-facto analysis by computing the Fréchet distance between a normalized driven path and a normalized "ideal" synthetic ground-truth path. This comparison of the driving trajectories, detailed in Appendix: A.5, were visually confirmed against the 360°-video recordings of participants, particularly in cases where the participants were clearly far off of the ideal path.

Self-reported Assessment We performed semi-structured interviews with participants to record their own impression of their ability to drive with the XR-OOM system. We asked people to describe their experience driving with the XR system and note differences between with and without it.

4.6 Risk Assessment & Mitigation

Real-world experiments can contain the risk of physical injury to the participant and other people and the risk of physical damage to the vehicle and the physical world. Our broader goal with the XR-OOM system is to run on real roads with regular traffic. Still, for this validation study, we exercised an abundance of caution to minimize possible sources of risk. A list of risk factors and our mitigation methods is given below:

4.6.1 Driver variability. We recruited drivers who had driver's licenses and were legally allowed to operate vehicles in the United States. The pre-study protocol involves some simple instructions for gaining consent and setting up driving activity. Participants who had difficulties with these steps would not have been asked to

continue with the study. We could have further restricted participation to people who had local driving experience and experience with automatic transmission vehicles; these restrictions trade-off against understanding issues that a broader diversity of participants would reveal, so we did not apply them to this study.

4.6.2 Unintended acceleration/collision with obstacles. We designed the activities to occur at low-speed and over short distances and located the tasks so that strong braking or accidental acceleration would be unlikely to result in a collision with any surrounding structures. The participants were asked to put on seatbelts early in the study, and those instructions were repeated at the start of every condition. We tested both physical and CAN-bus-based systems to limit the vehicle's speed. Subsequently, We found they were unnecessary and could introduce other safety issues because they cause the car to behave in ways that participants would not expect.

4.6.3 Collision/interference with other vehicles and pedestrians. We set our study in a restricted lot where we would not encounter wayward traffic. We had a permit for the exclusive use of the lot during our studies. The on-site researchers also directed people and vehicles driving through the site, so they did not come close to our test area. In advance, drivers were told that people drive through the site to just brake and wait until the other vehicles had passed through.

4.6.4 Researcher safety. The study was initially designed to occur with a researcher in the backseat of the vehicle that the participant was driving. To reduce the risk of viral transmission, we subsequently located the researcher outside the vehicle and provided participants with instructions and guidance over a mobile phone. Researchers made sure participants were fully stopped and asked them to put the car in park before approaching the vehicle and opening any doors. Researchers wore reflective traffic safety vests to increase their visibility and clearly stood at the same location throughout the study.

4.6.5 Loss of vision. Because the system uses a video passthrough, it is possible for hard failures in the XR system to cut driver visibility completely. In any on-road experiment, it is also possible that external obstacles and events can cause loss of vision. We told participants that in the unlikely case of complete loss of the ability to see the vehicle and driving site, they should immediately step on the brakes for a hard stop and await additional instructions.

4.6.6 Driver discomfort. Discomfort is a factor in many driving simulation studies, so much so that there are standardized questionnaires to assess the degree of "simulator sickness" participants experience [33]. Counter-intuitively, both Paredes et al. and Goedicke et al. found that, with a high-resolution virtual reality headset in a vehicle with well-correlated physical and virtual motion, the combination of the virtual reality environment and the physical environment causes *less* motion sickness than virtual reality environments or in-vehicle autonomous driving experiments themselves [19, 53]. This is probably because the correlation of virtual and physical motion induces fewer problems than the juxtaposition of moving environments without corresponding visual movement

(as commonly occurs in VR environments) or visually moving environments without corresponding physical motion (as occurs in driving simulation) [31].

While we would like not to have any discomfort, one thing we wish to understand is whether any experienced discomfort is tolerable or impairing. To minimize the likelihood of nausea and motion sickness, we had the air conditioning turned on in the car with the fan speed set to high. We also let the participants know that they could stop the study at any time if they felt uncomfortable. With mixed reality in a moving car, both motion sickness and simulator sickness need to be avoided [60]. Participant discomfort caused by simulation or motion sickness could further be mitigated by good ventilation and temperature control, which is a feature that can be expected in most on-road vehicles [60].

We did not include tasks that would involve reversing; the entire experiment takes place in the forward driving direction. This decision was based on two considerations: (1) rear driving tasks are already more challenging in normal driving conditions and thus include a higher risk, and (2) our system scope currently does not support a full rear driving experience due to virtual objects not appearing in mirrors inside the vehicle.

5 RESULTS

5.1 Task Measures

In the following sections, we discuss the extent to which participants are able to control the system to validate the XR-OOM system's usability as a simulation platform.

5.1.1 Cockpit Tasks. Cockpit task performance was evaluated as either success or failure. The data was extracted from the 360° camera in the same manner as for the driving tasks (the full table is in Appendix: A.3.1. Most participants could successfully conduct the tasks with minor retries in the headset condition. Some participants could not turn on the vehicle ($N=2$), used the wrong stalk for controlling the wipers ($N=1$), or looked for a button instead of operating the foot-controlled parking brake ($N=3$) in the headset condition. The cause of these errors mainly was related to unfamiliarity with the (type of) vehicle or the instructions, such as not knowing to press the brake when starting the vehicle or not knowing that the parking brake was foot-operated. The failed tasks did not rely heavily on visual information. In five of the tasks, some participants skipped the task due to the task already being completed (e.g., the seatbelt was fastened) - we did not include their counts in the total numbers of failed or completed tasks.

One notable exception was the last task in which the participant had to explain all the icons they could see on the dashboard. The vehicle speed (the largest number) was easily visible; however, smaller icons (e.g., tire pressure warning) could not be resolved. While some participants could see more than others, none could see and report all icons.

One notable difference during the study was that the majority of the participants needed to hold the instruction papers in front of the headset, contrary to reading directly from their lap or the center console when not wearing the headset (Condition A).

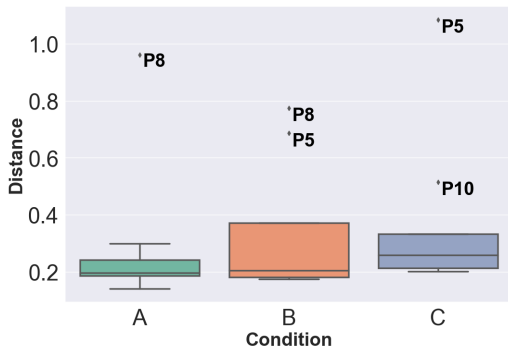


Figure 4: This box plot shows for each condition the Fréchet distance with the outliers showing the participants and conditions where the participants miss judged their path and followed the wrong route.

5.1.2 Driving Tasks. All participants could read the instruction diagrams and follow the routes. A human annotator extracted performance data about the participants driving by observing videos of the participant-facing camera, an external camera, and the headset camera. From these annotations, a set of tasks appeared to be particularly difficult. These are described below.

In the driving tasks, the Slalom task was the most difficult for participants to accomplish. Most participants needed to back up to do it correctly because the first turn into the slalom was very tight. This task was unforgiving; if participants missed the first turn, they would have to retrace their position to be correctly positioned for the subsequent task. The short height of the traffic cones made it challenging to see the cones when drivers got closer to them; participants had to remember the locations of the cones and guess where the vehicle’s bumpers were in space to avoid hitting the cones. This task was particularly challenging in condition C because the lack of ability to see the virtual cones in the rear-view mirror made it challenging to locate the relative location of those cones to the car.

Finding the correct point to turn right, for the *Left Turns* task was also hard for some participants. Several participants turned too early and then had a very long left turn.

Participants seemed to find the Stop Line task to be the easiest. However, the participant’s interpretation of where to stop for a stop sign line varied significantly.

5.2 Driving Trajectory

Each participant’s drive was recorded with ROS. However, 2 trajectory recordings were incomplete or corrupted after the experiment, which is why only data for 8 participants could be compared. From 8 recorded trajectories, we compute the Fréchet distances as a proxy for driving performance to compare the conditions against an “optimal” path. Please see Appendix: A.5 how these values were computed. The graph of the performance differences from these analyses (see Figure 4) The average Fréchet distance increase from condition A to C, though not significantly different.

5.3 Self-reported Measures

We also recorded each participant’s post-experiment interview results, including a self-assessment rating for each condition.

At the end of the experiment, participants were asked eight questions about their general experience and well-being. Three out of 10 participants mentioned they drove better with the headset. Three participants reported mild motion sickness, with one of those cases being more severe. Three people reported a light headache, and 4 participants reported some strain on their eyes. Although participants were asked to contact us if these symptoms failed to subside or if they got worse later in the day, no participant contacted us about these issues afterward.

5.4 System Performance

As part of our risk assessment and later to evaluate the system’s reliability, we distinguish between three different technical failure modes of the XR-System in conditions B and C:

5.4.1 HMD Blackout Due to Hardware Failure. No participants experienced any hardware failures in the validation study; all participants had continuous vision throughout all the headset conditions.

5.4.2 HMD Failure Due to a Software Failure. In one case, the headset calibration screen appeared; this screen should only occur when the headset is first put on. This failure was likely caused by an improperly worn headset and highlighted the importance of checks and calibrations for headset mounting early in the study. This type of failure is complex for the researcher to control. Still, in the future, we will alert participants to the possibility of this failure and give instructions on what to do (stop the vehicle by pressing on the brake) and how to prevent recurrence (adjust the headset so that it is mounted firmly on the head).

5.4.3 Loss of Tracking. Situations, where the headset’s location within the car was not correctly mapped, occurred several times during the study. We noticed that when some participants turned their heads to the left, towards the driver’s seat window, the in-car localization lost track of the headset. This was not a significant problem as tracking was resumed as soon as the participant turned their head back.

The in-world tracking was also unstable for some participants; in particular, variability in external lighting caused by cloud cover caused the tracking to be less reliable. The participants were always able to continue the experiment. The tracking loss resulted in small jumps (a few centimeters and degrees) in the visual field; no participants who experienced this had difficulty adjusting their trajectory due to the tracking jitter.

On the whole, and comparing the technical issues using this simulator to previous in-lab and in-vehicle driving simulation environments used by the study researchers, we felt this system was robust enough to perform driving experiments with.

6 DISCUSSION

As a driving simulation platform, XR-OOM needs to support a wide range of experimental studies. With the results from this implementation, we can now attempt to validate the system and contextualize it with other types of simulator studies and possible scenarios this kind of a system should ideally support.

6.1 System Validation

For system validation, we return to the questions first raised in Section 4.3. We should point out at the outset of this discussion that since what we are trying to do is validate the XR HMD setup for driving studies, the desired result is for there to be as little difference as possible between the performance between conditions. If there is a difference, we would prefer that difference to be small and the performance to be best in the no-headset condition. This would suggest that tasks people can perform well enough in the mixed reality setting would predict even better performance from the simulation. (This is different from what would be desirable in an experimental study, where low or no difference between conditions would indicate a null result.)

6.1.1 Does using the video pass-through headset impact the participants' ability and experience to operate cockpit tasks? Besides the cockpit task, we intentionally chose to have the instructions printed on paper for the participant to read to both ensure a correctly working HMD and evaluate the participant's ability to read and follow the instructions. All participants were capable of reading the cockpit tasks from the instruction sheet. So most of the cockpit tasks were also unaffected by the use of a headset (see Figure: 3). These results point to the system's exciting capability that allows for testing of brought-in devices like phones or tablets during the driving experience. These devices might be the participants' own real devices or virtual applications that do not yet exist.

The most problematic task was distinguishing icons on the dashboard. Camera resolution, dynamic range, and refresh rate aliasing made this task particularly difficult. While all participants could see some information, most only reported the vehicle speed, the largest number in the screen center. Additionally, it is essential to point out that seeing the media center and AC controls caused fewer difficulties; these, however, were not explicitly part of our study and need to be further assessed in future research.

Certain display/camera interference notwithstanding, this method appears to allow for testing of new displays and controls. With the present technology, it is possible to change any display surface. These should be done in a way that has face validity (simulated interfaces should strongly resemble actual interfaces) and elicit relatively realistic behavioral responses. This also includes heads-up displays and other new display technologies that go beyond current instrument clusters or center console media players.

Additional complications occurred with the tasks that required body movement in the vehicle, like setting up a chair and putting on the seat belt. Wearing a headset gave less space for the participant to move around and the limited field of view from the headset meant that many steps in these tasks needed to be done via touch (e.g., feeling where the seat belt is).

In general, the cockpit task did not pose any significant hurdle to running the study. For such tasks, further optimizations like, e.g., directly controlling the XR-camera's exposure could help make display items more visible.

6.1.2 Does using the video pass-through headset impact the participants' driving performance and experience? Based on the experiments, the difference of driving with a mixed reality headset vs. without one is minor. Variables like average speed and task time

are similar, and the vehicle paths appear to be similar when visually compared.

The main limitation identified during these trial runs was that the headset's weight and bulkiness caused participant discomfort. This, in combination with the limited space of the vehicle cabin, detracted from the participant experience.

These findings pertain specifically to low-speed driving tasks; we expect that driver performance and the driver's experience would degrade in higher speed conditions. However, based on the experiences we have had to date, we estimate that it should be possible to safely perform driving tasks up to 30 mph if the vehicle is in an area without other cars or pedestrians.

Even with the mentioned limitations, these findings enable this approach to test drivers' responses to different visual impairments to the driving environment (e.g., cataracts, heavy snow, or rain) and to design compensatory measures for such drivers. By altering the visual feed to the XR headset, we can simulate the impact visual impairments as limited field-of-view have on driving.

6.1.3 Are virtual objects simulated believably enough to provide valid predictions of how drivers will respond to physical objects in driving?

One key issue for this validity study is whether participants treat the virtual objects as they would a real object, enough so that their interactions with the virtual object indicate how they would interact with future physical objects. Realism of the simulation could be related to aspects such as depth perception in virtual displays, which has been studied in various automotive AR applications such as [6].

From study footage, we could see that participants drove more cautiously through the tasks with the virtual cones, taking a little more time and making a few more mistakes. In general, completing the tasks with condition C was more difficult than any of the other conditions. This is also reflected in the participants' self-reported experience.

Analysis of the driving paths for the three conditions (shown in Figure 4) shows that people tended to drive similar paths in all three conditions. This also indicates that the participants did not ignore or in any other way treat the virtual cones any different than the physical cones. After watching the headset video recordings and examining the generated path, the participants seem to take wider turns around the virtual cones, comparatively showing a more cautiousness when interacting with virtual objects.

This suggests that participants took the virtual objects seriously and tried not to hit the objects even though they were not physically present. For future studies, the traffic cones could be replaced by pedestrians, bicyclists, road incursions, road signs to be followed, etc. Additional mixed reality elements could then be introduced into the XR environment to accentuate the external phenomena (e.g., highlighting pedestrians it senses and recognizes) or feature simulated objects or actors to assess the participants' response to the highlights or simulated object.

The capability to meaningfully place and track objects in the environment also extends the capability of the simulator towards testing extraordinary traffic conditions like, e.g., an icy patch of road that affects user experience and vehicle response. While the current system cannot affect the control behavior of the car, it can visually augment the participant view to create a more holistic

experience in ways that a controlled laboratory or pure VR studies would not be able to.

6.1.4 Does the system capture information about participant behavior well enough for analysis in user studies? In the analysis above, we highlighted some of the available data streams we captured and demonstrated how they could be used to gain insight into a participants' behavior and performance.

The XR-OOM system offers multiple perspectives for qualitative coding that researchers can use to observe a participant post-facto. Video streams provide a clear view of the vehicle, participant, the road in front of the vehicle and directly show the participant's perspective and track their gaze. These streams can also be automatically analyzed as shown in Appendix: A.4. These different perspectives allow the researcher to reconstruct the complete picture of what the participant saw and how they reacted in a given situation.

Capturing detailed participants' responses is key when trying to understand their reaction to a given social environment; since these have shown to have an enormous impact on drivers both good and bad. The combination of XR-OOM capabilities to collect detailed data and render virtual objects allows for simulated social actors in the car. By adding sound and other virtual characters into the vehicle, we could evaluate how social distractions affect a driver and how technology could help mitigate such effects. Conversely, social agents could prevent drivers from becoming bored or angry in the vehicle or help point out important external cues to improve situation awareness.

Further, qualitative data can be gathered through self-reported questionnaires and interviews. This study used a regular phone connection to allow the participant to think aloud as they worked through the different tasks and answer questionnaires at the end of each condition. This is made possible by the headset's high resolution as it resolves the phone screen clearly enough that the participant can use their phone.

The quantitative data allows more a formative analysis of the participants' driving. Sensors like the LIDAR or IMU will enable us to capture the detailed movements of the research vehicle and compute an accurate path that can highlight the driving behavior.

The combination of both participant behavior and vehicle motion are vital features to test advances in vehicle automation. XR-OOM could support the study and development of different paths to driving automation, such as teleoperation [49] or supervisory control [30]. For such studies, relative behavioral validity in the way that people notice and respond to control transition events is desirable [28, 48].

6.1.5 Other Discovered Issues. Beyond answering the questions we posed about the viability of XR-OOM as a driving simulation environment, our research also helped us discover technical issues that should be addressed, particularly regarding the cameras used on the mixed reality headset.

Because mixed reality equipment is often designed to operate indoors and under controlled lighting conditions, the cameras on these systems are not well-optimized to handle the high-dynamic-range of light intensity in an outdoor setting. This can especially be problematic inside the research vehicle, as participants might switch between looking at the outside world and a shaded dashboard.

Another issue we encountered is the interaction between the refresh rates of the camera and the instrument panels the participant is trying to read from. If these have similar refresh rates, aliasing can occur; this makes the panel displays appear to fade in and out and thus become harder to read.

6.2 Safety Assessment and Risk Mitigation

For safety assessment and risk mitigation, we return to the risks we previously identified in Section 4.6. We also discuss the necessity and efficacy of the measures we took for risk reduction in this validation work.

6.2.1 Driver Variability. The participants had different comfort levels and capabilities accomplishing the cockpit and driving tasks, even for the control condition tasks. Some participants were flustered and inadvertently skipped tasks. Another source of variability came from the participants' experience driving a vehicle like the research vehicle, e.g., some participants had only driven manual transmission vehicles and therefore took longer to find and use the parking brake. The participants also had differing responses to the XR headset: their differences in height and neck strength affected their response to the system. While we were prepared to reject participants who seemed like they might not perform the study tasks safely, we did not find that we needed to stop the study early for any of the participants in this study.

6.2.2 Unintended Acceleration/Collision with Obstacles. The task design was successful at limiting the participant's speed as the recorded maximum was only 4.3 m/s. At no point did any safety systems like speed limiters, verbal collision warnings, or safety belts engage.

6.2.3 Collision/interference with Other Vehicles and Pedestrians. In some cases, other vehicles or pedestrians drove through the lot. At no point did it seem like the participants were likely to collide with any other vehicles or pedestrians; additionally, the participant either noticed the other vehicles or passers-by themselves or was notified by the researcher over the audio channel to pause study driving activity.

6.2.4 Researcher Safety. The researchers had a clear spot to the side of the experiment area and used clear communication and to confirm parking with the participants when they approached the vehicle between conditions.

6.2.5 Loss of Vision. Because the system uses a video passthrough, it is possible for hard failures in the XR system to cut driver visibility completely. (These are discussed in Section 5.4). While we did not have any complete HMD blackouts, we did have an instance where the HMD popped dialog boxes on the display, which blocked the areas of the participant's visual field. In that instance, the driver was able to stop and follow audio instructions to resolve the issue. In the future, it is important to find ways to disable such messages unless the vehicle is completely stopped. However, the possibility of this type of failure leads us to recommend that driving with HMD should be restricted to low-speed (<15 mph) driving scenarios in real-world environments and mid-range speeds (<30 mph) in closed track or black-lake environments. Higher speed driving should

employ XR systems with optical pass-through, such as the Microsoft HoloLens[45].

6.2.6 Driver discomfort. While we anticipated and hoped to mitigate discomfort stemming from simulator or motion sickness, what we found instead was the major source of driver discomfort in the study came from the headset, which many participants found to be heavy. Prior research has indicated that heavy or unbalanced headsets can create discomfort in participants over time [37, 66]. We hope that future headsets are lighter and less likely to cause this issue.

Across the study conditions, condition C was rated the least comfortable, but it was still above the mid-point for the Likert (2.6 compared to 2.5 out of 5). We also asked participants to email us if they felt nausea later in the day following the study, and none of the participants sent an email indicating post-experiment discomfort.

6.3 Limitations

6.3.1 COVID-19 Protocol Measures. While this system was developed before the COVID-19 pandemic, it was evaluated during the period following strict social distancing measures. Our study protocol included specific transmission prevention measures not detailed above: Participants and researchers were required to wear face masks throughout the study. The headset and research vehicle were extensively sanitized before and after each participant run. The research vehicle window was left open to enable extra ventilation. Participant recruitment was limited to the on-campus population who were part of an extensive university testing program.

We do not indicate that these preventative measures impacted the study results, and all of these measures should have affected all of our study conditions equally.

6.3.2 Convenience Sampling. Due to the aforementioned study restrictions, our study participant pool was limited to part of an on-campus population. Ideally, this work would be replicated with a broader demographic pool to see if age or other factors affect the system function.

6.3.3 Reverse Driving Tasks. Participants in our study were able to conduct short stints of backing up the vehicle, but on the whole, were not asked to perform reverse driving tasks. Because driving in reverse would require a model of the world behind the vehicle, instrumentation of the rear-view mirror, and complex graphical modeling, this system was not validated for tasks that would require extensive backward driving, such as parallel parking maneuvers. This is a limitation common to many driving simulators. The scope of our XR-OOM platform is focused on forward driving tasks.

6.3.4 Other Road Users. This current paper's contribution focuses on the system design and validation of the safety and usability of the system for driving simulation experiments. This was critical to establish before attempting to use XR-OOM in environments with other drivers, bicyclists, and pedestrians. Additional safety and validation research should be done before using this system for driving simulation in an environment with other road users.

6.3.5 Order and Carryover Effects. In our experiments, we discovered carry-over effects in the driving conditions (A without headset,

B with headset with pass-through video, C with virtual headset objects).

Specifically, we saw that when task A is run first, tasks A and C are ranked lower across all metrics. When task A is run after task B or C, it is rated higher across all metrics. Task B is rated higher when done after task C, and task C is ranked higher in control when done first. Similar effects can be seen when we analyze the effect of the second task run. In the last task analysis, all tasks are rated lower across metrics when task C is the last task performed. This implies that, other tasks are perceived as easier after performing task C, and task C is perceived as harder when run after the other tasks.

This is not a concern for our system validation test or for experiments that take place in a headset condition only, but should be kept in mind when the XR-OOM system is being used for comparing between different headset conditions.

7 CONCLUSION

By mixing virtual reality simulation with on-road and in-vehicle driving, we enable human-machine interaction researchers and designers to develop and test future interfaces and possible driving interactions within quasi-real world environments. While several research groups are working on such a system, this paper is the first to describe in detail how mixed reality technology needs to be explicitly designed to enable and instrument driving simulation studies. Furthermore, we have taken the first steps to validate that such a system is safe to operate and can be used to predict the performance and behavior of real-world driving.

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System diagrams, code, and simulator data for XR-OOM will be made available on GitHub (<https://github.com/far-lab/XR-OOM> and <https://github.com/far-lab/xr-oom-localization>) under the MIT open-source license at publication time.

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REFERENCES

- [1] Talal Al-Shihabi and Ronald R Mourant. 2003. Toward more realistic driving behavior models for autonomous vehicles in driving simulators. *Transportation Research Record* 1843, 1 (2003), 41–49.
- [2] R Wade Allen, Theodore J Rosenthal, and Marcia L Cook. 2011. *A short history of driving simulation*. CRC press, Boca Raton, Chapter 2, 2–1 to 2–16.
- [3] Ignacio Alvarez, Laura Rumbel, and Robert Adams. 2015. Skyline: a rapid prototyping driving simulator for user experience. In *Proceedings of the 7th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. ACM, 101–108.
- [4] Kaarin J. Anstey, Joanne Wood, Stephen Lord, and Janine G. Walker. 2005. Cognitive, sensory and physical factors enabling driving safety in older adults. *Clinical Psychology Review* 25, 1 (2005), 45 – 65. <https://doi.org/10.1016/j.cpr.2004.07.008>
- [5] Sonia Baltodano, Srinath Sibi, Nikolas Martelaro, Nikhil Gowda, and Wendy Ju. 2015. The RRADS Platform: A Real Road Autonomous Driving Simulator. In *Proceedings of the 7th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (Nottingham, United Kingdom) (*AutomotiveUI '15*). Association for Computing Machinery, New York, NY, USA, 281–288. <https://doi.org/10.1145/2799250.2799288>
- [6] Alexandra W.D. Bremers, Ali Özgür Yöntem, Kun Li, Daping Chu, Valerian Meijering, and Christian P. Janssen. 2021. Perception of perspective in augmented

- reality head-up displays. *International Journal of Human-Computer Studies* 155 (2021), 102693. <https://doi.org/10.1016/j.ijhcs.2021.102693>
- [7] Heiner Bubb, Klaus Bengler, Rainer E. Grünen, and Mark Vollrath. 2015. *Automobilergonomie*. Springer Fachmedien Wiesbaden, Wiesbaden.
 - [8] GE Burnett, A Irune, and A Mowforth. 2007. Driving simulator sickness and validity: How important is it to use real car cabins. *Advances in transportation studies* 13 (2007), 33–42.
 - [9] Volvo Cars. 2019. Volvo Cars & Varjo – Design and safety development in mixed reality. [YouTube video]. Retrieved January 8, 2022 from https://www.youtube.com/watch?v=tCv0hJGBo_I
 - [10] Mark Claypool and Kaja Claypool. 2006. Latency and Player Actions in Online Games. *Commun. ACM* 49, 11 (Nov. 2006), 40–45. <https://doi.org/10.1145/1167838.1167860>
 - [11] State Farm Mutual Automobile Insurance Company. unknown date. *Teen driving 101: a step-by-step test of essential driving skills*. State Farm Mutual Automobile Insurance Company. Retrieved January 10, 2022 from <https://www.statefarm.com/simple-insights/auto-and-vehicles/teen-driving-101-a-stepbystep-test-of-essential-skills>
 - [12] Eduardo Cuervo, Krishna Chintalapudi, and Manikanta Kotaru. 2018. Creating the Perfect Illusion: What Will It Take to Create Life-Like Virtual Reality Headsets?. In *Proceedings of the 19th International Workshop on Mobile Computing Systems & Applications* (Tempe, Arizona, USA) (*HotMobile '18*). Association for Computing Machinery, New York, NY, USA, 7–12. <https://doi.org/10.1145/3177102.3177115>
 - [13] Forward Development. 2017. City Car Driving [PC Game]. Version 1.5. <http://www.citycardriving.com/>
 - [14] Tilman Dinger, Kai Kunze, and Benjamin Outram. 2018. VR Reading UIs: Assessing Text Parameters for Reading in VR. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (*CHI EA '18*). Association for Computing Machinery, New York, NY, USA, 1–6. <https://doi.org/10.1145/3170427.3188695>
 - [15] Donald L Fisher, Jeff K Caird, Matthew Rizzo, and John D Lee. 2011. Handbook of driving simulation for engineering, medicine and psychology: an overview. *Handbook of driving simulation for engineering, medicine, and psychology* (2011), 1–1 to 1–14.
 - [16] Frank Flemisch, Matthias Heesen, Tobias Hesse, Johann Kelsch, Anna Schieben, and Johannes Beller. 2012. Towards a dynamic balance between humans and automation: authority, ability, responsibility and control in shared and cooperative control situations. *Cognition, Technology & Work* 14, 1 (2012), 3–18.
 - [17] G Geiser. 1985. Man-machine communication in motor vehicles.
 - [18] Florin-Timotei Ghiurău, Mehmet Aydın Baytaş, and Casper Wickman. 2020. ARCAR: On-Road Driving in Mixed Reality by Volvo Cars. In *Adjunct Publication of the 33rd Annual ACM Symposium on User Interface Software and Technology*. Association for Computing Machinery, 62–64.
 - [19] David Goedicke, Jany Li, Vanessa Evers, and Wendy Ju. 2018. *VR-OOM: Virtual Reality On-Road Driving Simulation*. Association for Computing Machinery, New York, NY, USA, 1–11. <https://doi.org/10.1145/3173574.3173739>
 - [20] David Goedicke, Jany Li, Vanessa Evers, and Wendy Ju. 2018. *VR-OOM: Virtual Reality On-Road Driving Simulation*. Association for Computing Machinery, New York, NY, USA, 1–11. <https://doi.org/10.1145/3173574.3173739>
 - [21] Christian Gold, Lutz Lorenz, Daniel Damböck, and Klaus Bengler. 2013. Partially automated driving as a fallback level of high automation. *Tagung Fahrerassistenzsysteme. Der Weg zum automatischen Fahren* 28, 29.11 (2013), 2013.
 - [22] Lauren Goode. 2019. Get Ready to Hear a Lot More About 'XR'. Retrieved January 8, 2022 from <https://www.wired.com/story/what-is-xr/>
 - [23] Peter Grant, Bruce Artz, Jeff Greenberg, and Larry Cathey. 2001. Motion characteristics of the VIRTTEX motion system. In *Proceedings of the 1st human-centered transportation simulation conference*. 4–7.
 - [24] Jeffrey Allen Greenberg and Mike Blommer. 2011. *Physical fidelity of driving simulators*. CRC press, Boca Raton, 7–1 to 7–24.
 - [25] Cameron Grout, William Rogers, Mark Apperley, and Steve Jones. 2015. Reading Text in an Immersive Head-Mounted Display: An Investigation into Displaying Desktop Interfaces in a 3D Virtual Environment. In *Proceedings of the 15th New Zealand Conference on Human-Computer Interaction* (Hamilton, New Zealand) (*CHINZ 2015*). Association for Computing Machinery, New York, NY, USA, 9–16. <https://doi.org/10.1145/2808047.2808055>
 - [26] Peter A Hancock and Thomas B Sheridan. 2011. *The future of driving simulation*. CRC press, Boca Raton, 4–1 to 4–11.
 - [27] Rebecca L Hartman and Marilyn A Huestis. 2013. Cannabis effects on driving skills. *Clinical chemistry* 59, 3 (2013), 478–492.
 - [28] Philipp Hock, Johannes Kraus, Franziska Babel, Marcel Walch, Enrico Rukzio, and Martin Baumann. 2018. How to design valid simulator studies for investigating user experience in automated driving: review and hands-on considerations. In *Proceedings of the 10th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. Association for Computing Machinery, 105–117.
 - [29] F. Hu, Y. Deng, W. Saad, M. Bennis, and A. H. Aghvami. 2020. Cellular-Connected Wireless Virtual Reality: Requirements, Challenges, and Solutions. *IEEE Communications Magazine* 58, 5 (2020), 105–111.
 - [30] Karl Iagnemma. 2019. Supervisory control of vehicles. US Patent 10,303,166.
 - [31] Quinate Chioma Ihemedu-Steinke, Stanislava Rangelova, Michael Weber, Rainer Erbach, Gerrit Meixner, and Nicola Marsden. 2017. Simulation sickness related to virtual reality driving simulation. In *International Conference on Virtual, Augmented and Mixed Reality*. Springer, Springer International Publishing, 521–532.
 - [32] Robert JK Jacob, Audrey Girouard, Leanne M Hirshfield, Michael S Horn, Orit Shaer, Erin Treacy Solovey, and Jamie Zigelbaum. 2008. Reality-based interaction: a framework for post-WIMP interfaces. In *Proceedings of the SIGCHI conference on Human factors in computing systems*. ACM, 201–210.
 - [33] Robert S Kennedy, Norman E Lane, Kevin S Berbaum, and Michael G Lilienthal. 1993. Simulator sickness questionnaire: An enhanced method for quantifying simulator sickness. *The international journal of aviation psychology* 3, 3 (1993), 203–220.
 - [34] Dagmar Kern and Albrecht Schmidt. 2009. Design Space for Driver-Based Automotive User Interfaces. In *Proceedings of the 1st International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (Essen, Germany) (*AutomotiveUI '09*). Association for Computing Machinery, New York, NY, USA, 3–10. <https://doi.org/10.1145/1620509.1620511>
 - [35] Esko Keskinen, Mika Hatakka, Sirkku Laapotti, Ari Katila, and M Peräaho. 2004. Driver behaviour as a hierarchical system. *Traffic and Transport Psychology* (2004), 9–29.
 - [36] Advanced Realtime Tracking GmbH & Co. KG. 2019. *SmartTrack3*. Advanced Realtime Tracking GmbH & Co. KG. Retrieved January 8, 2022 from <https://ar-tracking.com/products/tracking-systems/smarttrack3/>
 - [37] Eunjee Kim and Gwanseob Shin. 2020. User discomfort while using a virtual reality headset as a personal viewing system for text-intensive office tasks. *Ergonomics* 64, 7 (2020), 1–9.
 - [38] Lars Knaack, Ann-Karolin Lache, Oliver Preikszas, Sascha Reinhold, and Michael Teistler. 2019. Improving Readability of Text in Realistic Virtual Reality Scenarios: Visual Magnification Without Restricting User Interactions. In *Proceedings of Mensch Und Computer 2019* (Hamburg, Germany) (*MuC'19*). Association for Computing Machinery, New York, NY, USA, 749–753. <https://doi.org/10.1145/3340764.3344902>
 - [39] Kenji Koide, Jun Miura, and Emanuele Menegatti. 2019. A portable three-dimensional LIDAR-based system for long-term and wide-area people behavior measurement. *International Journal of Advanced Robotic Systems* 16 (02 2019). <https://doi.org/10.1177/1729881419841532>
 - [40] Ruilin Liu, Daehan Kwak, Srinivas Devarakonda, Kostas Bekris, and Liviu Ifteod. 2017. Investigating Remote Driving over the LTE Network. In *Proceedings of the 9th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (Oldenburg, Germany) (*AutomotiveUI '17*). Association for Computing Machinery, New York, NY, USA, 264–269. <https://doi.org/10.1145/3122986.3123008>
 - [41] Katerina Mania, Bernard D. Adelstein, Stephen R. Ellis, and Michael I. Hill. 2004. Perceptual Sensitivity to Head Tracking Latency in Virtual Environments with Varying Degrees of Scene Complexity. In *Proceedings of the 1st Symposium on Applied Perception in Graphics and Visualization* (Los Angeles, California, USA) (*APGV '04*). Association for Computing Machinery, New York, NY, USA, 39–47. <https://doi.org/10.1145/1012551.1012559>
 - [42] A James McKnight and Bert B Adams. 1970. Driver Education Task Analysis. Volume I: Task Descriptions. Final Report (August 1969-July 1970). (1970).
 - [43] Natasha Merat, A Hamish Jamson, Frank CH Lai, Michael Daly, and Oliver MJ Carsten. 2014. Transition to manual: Driver behaviour when resuming control from a highly automated vehicle. *Transportation research part F: traffic psychology and behaviour* 27 (2014), 274–282.
 - [44] John A. Michon. 1985. *A Critical View of Driver Behavior Models: What Do We Know, What Should We Do?* Springer US, Boston, MA, 485–524. https://doi.org/10.1007/978-1-4613-2173-6_19
 - [45] Evan Miller, Bee Hanson, Diana Hanson, and Yashar Bahman. 2021. *Moving Platform Mode on Low Dynamic Motion Moving Platforms*. Microsoft. Retrieved January 8, 2022 from <https://docs.microsoft.com/en-us/hololens/hololens2-moving-platform>
 - [46] Brian Ka-Jun Mok, David Sirkin, Srinath Sibi, David Bryan Miller, and Wendy Ju. 2015. Understanding driver-automated vehicle interactions through Wizard of Oz design improvisation. In *Proceedings of the International Driving Symposium on Human Factors in Driver Assessment, Training and Vehicle Design*. 386–392.
 - [47] Takahiko Murano, Takashi Yonekawa, Masami Aga, and Sueharu Nagiri. 2009. Development of high-performance driving simulator. *SAE International Journal of Passenger Cars-Mechanical Systems* 2, 2009-01-0450 (2009), 661–669.
 - [48] Frederik Naujoks, Sebastian Hergeth, Katharina Wiedemann, Nadja Schömig, Yannick Forster, and Andreas Keinath. 2019. Test procedure for evaluating the human-machine interface of vehicles with automated driving systems. *Traffic injury prevention* 20, sup1 (2019), S146–S151.
 - [49] Stefan Neumeier, Nicolas Gay, Clemens Dannhenn, and Christian Facchi. 2018. On the way to autonomous vehicles teleoperated driving. In *AmE 2018-Automotive meets Electronics; 9th GMM-Symposium*. VDE, VDE, 1–6.
 - [50] Stefan Neumeier, Philipp Wintersberger, Anna-Katharina Frison, Armin Becher, Christian Facchi, and Andreas Riemers. 2019. Teleoperation: The Holy Grail to

- Solve Problems of Automated Driving? Sure, but Latency Matters. In *Proceedings of the 11th International Conference on Automotive User Interfaces and Interactive Vehicular Applications* (Utrecht, Netherlands) (*AutomotiveUI '19*). Association for Computing Machinery, New York, NY, USA, 186–197. <https://doi.org/10.1145/3342197.3344534>
- [51] Cynthia Owsley and Gerald McGwin Jr. 1999. Vision impairment and driving. *Survey of ophthalmology* 43, 6 (1999), 535–550.
- [52] Lothar Pantel and Lars C. Wolf. 2002. On the Impact of Delay on Real-Time Multiplayer Games. In *Proceedings of the 12th International Workshop on Network and Operating Systems Support for Digital Audio and Video* (Miami, Florida, USA) (NOSSDAV '02). Association for Computing Machinery, New York, NY, USA, 23–29. <https://doi.org/10.1145/507670.507674>
- [53] Pablo E Paredes, Stephanie Balters, Kyle Qian, Elizabeth L Murnane, Francisco Ordóñez, Wendy Ju, and James A Landay. 2018. Driving with the fishes: Towards calming and mindful virtual reality experiences for the car. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 2, 4 (2018), 1–21.
- [54] Klaus Petersen. 2021. AR HMD for In-Car applications - LPVIZ (part 1) - LP-RESEARCH. Retrieved January 8, 2022 from <https://lp-research.com/ar-hmd-for-in-car-applications/>
- [55] Gill Pratt. 2017. *Self-Driving Cars: Road to Deployment*. Toyota Research Institute. Retrieved January 8, 2022 from <https://energycommerce.house.gov/sites/democrats.energycommerce.house.gov/files/Testimony-Pratt-DCCP-Hrg-Self-Driving-Cars-2017-02-14.pdf>
- [56] Morgan Quigley, Ken Conley, Brian Gerkey, Josh Faust, Tully Foote, Jeremy Leibs, Rob Wheeler, Andrew Y Ng, et al. 2009. ROS: an open-source Robot Operating System. In *ICRA workshop on open source software*. Kobe, Japan, 5.
- [57] J. Rasmussen. 1983. Skills, rules, and knowledge; signals, signs, and symbols, and other distinctions in human performance models. *IEEE Transactions on Systems, Man, and Cybernetics* SMC-13, 3 (1983), 257–266.
- [58] Volvo Cars (Press Release). 2019. Using augmented and virtual reality for the best volvo cars possible. <https://group.volvocars.com/news/future-mobility/2019/varjo-collaboration>
- [59] Anna Schieben, Matthias Heesen, Julian Schindler, Johann Kelsch, and Frank Flemisch. 2009. The theater-system technique: Agile designing and testing of system behavior and interaction, applied to highly automated vehicles. In *Proceedings of the 1st International Conference on Automotive User Interfaces and Interactive Vehicular Applications*. ACM, 43–46.
- [60] Joseph Smyth, Jonathan Robinson, Rebecca Burridge, Paul Jennings, and Roger Woodman. 2021. Towards the Management and Mitigation of Motion Sickness—An Update to the Field. In *Congress of the International Ergonomics Association*. Springer, Springer International Publishing, 834–840.
- [61] Inc. StereoLabs. 2019. *Meet ZED 2*. StereoLabs, Inc. Retrieved January 8, 2022 from <https://www.stereolabs.com/zed-2/>
- [62] M. Tonniss, V. Broy, and G. Klinker. 2006. A Survey of Challenges Related to the Design of 3D User Interfaces for Car Drivers. In *3D User Interfaces (3DUI'06)*. IEEE Computer Society, 127–134. <https://doi.org/10.1109/VR.2006.19>
- [63] Jan BF Van Erp and Pieter Padmos. 2003. Image parameters for driving with indirect viewing systems. *Ergonomics* 46, 15 (2003), 1471–1499.
- [64] Volvo. 2019. AWE USA 2019 - Exhibitor, Varjo Technologies. <https://www.awexr.com/usa-2019/exhibitor/2722-varjo-technologies>
- [65] Peter Wang, Srinath Sibi, Brian Mok, and Wendy Ju. 2017. Marionette: Enabling On-Road Wizard-of-Oz Autonomous Driving Studies. In *Proceedings of the 2017 ACM/IEEE International Conference on Human-Robot Interaction*. ACM, 234–243.
- [66] Yan Yan, Ke Chen, Yu Xie, Yiming Song, and Yonghong Liu. 2018. The effects of weight on comfort of virtual reality devices. In *International Conference on Applied Human Factors and Ergonomics*. Springer, Springer International Publishing, 239–248.
- [67] Dohyeon Yeo, Gwangbin Kim, and SeungJun Kim. 2019. MAXIM: Mixed-reality Automotive Driving XIMulation. In *2019 IEEE International Symposium on Mixed and Augmented Reality Adjunct (ISMAR-Adjunct)*. IEEE, 460–464. <https://doi.org/10.1109/ISMAR-Adjunct.2019.00124>